Ahatsham Hayat

Postdoctoral Research Fellow, Department of Anesthesia – Research
Harvard Medical School, Harvard University, USA

Preferred Email: afnu@bwh.harvard.edu   Preferred Phone: (+1) 531-229-9405
ahatsham.github.io   Google Scholar
Local Address: 2 Oakland Street Apt. 2, Brighton, MA 02135, USA

Research Interests and Lab Work

My research focuses on developing generalizable, reliable, and clinically meaningful multimodal AI systems for complex longitudinal data. My work spans large language models, vision-language models, NLP, time-series forecasting, multimodal learning, computer vision, and behavioral modeling, with emphasis on models that remain robust across populations, cohorts, and distribution shifts.

In Dr. Vesela Kovacheva's lab, I will develop predictive models to improve maternal patient safety and prevent adverse pregnancy outcomes. This work will use multidimensional clinical data, including ECG/EEG waveform signals, structured EHR data, genetic information, imaging, and clinical text, to build NLP-based, time-series, and multimodal models for predicting hypertensive crises, hemodynamic instability, hemorrhage, and ICU admission. The goal is to create accurate, interpretable, and clinically actionable AI systems for integration into clinical practice.

I have authored **35 peer-reviewed publications** and have experience working on **NSF, NIH**, and collaborative research grants.

Education

Ph.D. in Computer Engineering — Dates Attended: 2023–2026
University of Nebraska–Lincoln, Lincoln, Nebraska, USA
Completed: 2026   GPA: 4.0/4.0
Thesis: *A Multimodal Approach for Building Generalizable and Reliable Models of Longitudinal Experiential Data*

M.S. in Telecom Engineering — Dates Attended: 2025–2025
University of Nebraska–Lincoln, Lincoln, Nebraska, USA
Completed: 2025   GPA: 4.0/4.0

M.Tech in Computer Science — Dates Attended: 2014–2016
Visvesvaraya National Institute of Technology (VNIT), India
Completed: 2016   GPA: 7.5/10
Thesis: *Stream Clustering Algorithm for Detecting Temporal Contexts*

B.Tech in Computer Science — Dates Attended: 2010–2014
Uttarakhand Technical University, India

Completed: 2014 Grade: 70%
Thesis: *Fake News Detection using Binary Classification*

Work Experience

Postdoctoral Research Fellow June 2026 – Present
Brigham and Women's Hospital (BWH), Department of Anesthesia – Research, Harvard Medical School, USA

- Develop predictive models to enhance maternal patient safety and prevent adverse pregnancy outcomes.
- Analyze multidimensional clinical datasets, including ECG/EEG waveform signals, imaging, genetic data, structured EHR data, and clinical text.
- Build NLP-based, time-series, and multimodal learning models for risk prediction of hypertensive crises, hemodynamic instability, hemorrhage, and ICU admission.
- Contribute to clinically actionable AI systems designed for integration into real-world clinical practice.

Graduate Research Assistant Jan 2023 – May 2023
University of Nebraska–Lincoln, USA

- Lead researcher on NSF-funded projects in AI-enabled educational feedback and behavioral modeling.
- Developed multimodal architectures integrating LLMs, VLMs, and time-series models.
- Created narrative-based LLM pipelines for cross-cohort generalization.
- Designed automated LLM educational-feedback and behavioral reasoning systems.

Research Assistant Nov 2021 – Jan 2023
Madeira Interactive Technologies Institute (MITI), Portugal

- Built computer-vision models and sensing systems for agricultural applications (BASE Project).
- Developed banana yield prediction and automated harvesting support systems.

Assistant Professor (Research) Nov 2017 – Aug 2021
University of Petroleum and Energy Studies (UPES), India

- Taught core and advanced courses in machine learning, deep learning, AI, and algorithms. Complete course list available in the **Teaching Experience** section.
- Supervised undergraduate and postgraduate thesis projects.
- Published research in applied machine learning and computer vision.

Assistant Professor (Research) July 2016 – Nov 2017
Parul University, India

- Taught foundational courses in programming, data structures, algorithms, and introductory AI. Full teaching record listed in the **Teaching Experience** section.
- Supervised undergraduate research and contributed to curriculum development.

- Mentored students in applied ML and introductory computer vision.

Teaching Experience

Fall 2016–2017	Operating Systems; Introduction to Artificial Intelligence (Parul University, India)
Spring 2016–2017	Introduction to Object-Oriented Programming (Parul University, India)
Fall 2017–2018	Introduction to Machine Learning (University of Petroleum and Energy Studies, India)
Spring 2017–2018	Data Structures & Algorithms; Introduction to Operating Systems (University of Petroleum and Energy Studies, India)
Fall 2018–2019	Introduction to Machine Learning; Automata Theory (University of Petroleum and Energy Studies, India)
Spring 2018–2019	Data Structures & Algorithms (University of Petroleum and Energy Studies, India)
Fall 2019–2020	Introduction to Machine Learning; Introduction to Deep Learning (University of Petroleum and Energy Studies, India)
Spring 2019–2020	Introduction to Artificial Intelligence; Data Structures & Algorithms (University of Petroleum and Energy Studies, India)
Spring 2020–2021	Introduction to Machine Learning; Introduction to Deep Learning (University of Petroleum and Energy Studies, India)
Fall 2020–2021	Introduction to Artificial Intelligence; Data Structures & Algorithms (University of Petroleum and Energy Studies, India)

Student Mentorship

- **Hunter Tridle**
  B.S. in Electrical and Computer Engineering, University of Nebraska–Lincoln
  Mentored: Summer 2025 – Fall 2025
- **Helen Martinez**
  B.S. in Mathematics, University of Nebraska–Lincoln
  Mentored: Fall 2024 – Spring 2025
- **Sharif Akil**
  B.S. in Electrical and Computer Engineering, University of Nebraska–Lincoln
  Mentored: Fall 2023 – Spring 2024
- **Nathan Roberts**
  B.S. in Electrical and Computer Engineering, University of Nebraska–Lincoln
  Mentored: Spring 2023

Books Edited

Industry 5.0: Emerging Paradigms for Business, Technology, and Society

CRC Press, 2025
Editors: D. Sharma, M. Gupta, R. K. Kovid, T. P. Singh, P. Kumar, **A. Hayat**

Publications

2026 (Submitted)

1. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *Cross-Distribution Generalization in Longitudinal Behavioral Data Through Frozen Coherence Constraints*, Submitted to *Conference on Neural Information Processing Systems (NeurIPS 2026).*

2. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *Cross-Cohort Generalization in Behavioral Data Requires Semantic Anchoring*, Submitted to *Conference on Neural Information Processing Systems Position Paper Track (NeurIPS 2026).*

2026

3. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *Complementing Text Serialization with Visual Encoding of Structured Temporal Data for Cross-Distribution Generalization*, *Conference on Empirical Methods in Natural Language Processing (EMNLP Findings 2026).*

4. H. A. Diefes-Dux, **Hayat Ahatsham**, M. R. Hasan, *WIP: AI Tool for Theory-Aligned Grading and Feedback of Student Reflections, Preliminary Findings for Monitoring*, *IEEE*, October 2026.

5. **Ahatsham Hayat**, Neeta Kantamneni, Bilal Khan, Mohammad Rashedul Hasan, *Human-Guided Generative AI for Proactive STEM Student Support*, *The 2026 ASEE Annual Conference.*

6. **Ahatsham Hayat**, Hunter Tridle, Mohammad Rashedul Hasan, *From Numbers to Narratives: Efficient Language Model-Based Detection for Safety-Critical Minority Classes*, *EACL Findings 2026.*

2025

7. **Ahatsham Hayat**, Bilal Khan, Mohammad Rashedul Hasan, *ConText-LE: Cross-Distribution Generalization for Longitudinal Experiential Data via Narrative-Based LLM Representations*, *Conference on Empirical Methods in Natural Language Processing (EMNLP 2025) Findings.*

8. **Ahatsham Hayat**, Helen Martinez, Bilal Khan, Mohammad Rashedul Hasan, *A Three-Tier LLM Framework for Forecasting Student Engagement from Qualitative Longitudinal Data*, *The 29th Conference on Computational Natural Language Learning (CoNLL 2025).*

9. **Ahatsham Hayat**, Helen Martinez, Bilal Khan, Mohammad Rashedul Hasan, *Harnessing Language Models to Predict and Enhance STEM Engagement Using Non-Cognitive Experiential Data*, *The 2025 ASEE Annual Conference.*

10. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *A Context-Aware Approach for Enhancing Data Imputation with Pre-trained Language Models*, *The 31st International Conference on Computational Linguistics (COLING 2025).*

11. A. Haleem, M. Javaid, **Ahatsham Hayat**, R. Suman, S. Rab, R. P. Singh, *Cloud computing applications for Industry 4.0: A literature-based study*, *Journal of Industrial Integration and Management*, World Scientific Publishing Company.

12. P. Baglat, **Ahatsham Hayat**, S. S. Mostafa, F. Mendonça, F. Morgado-Dias, *Comparative analysis and evaluation of YOLO generations for banana bunch detection*, *Smart Agricultural Technology*, Volume 12, 2025, 101100, ISSN 2772-3755.

2024

13. **Ahatsham Hayat**, Bilal Khan, Mohammad Rashedul Hasan, *Leveraging Language Models for Analyzing Experiential Time-Series Data in Education*, *International Conference on Machine Learning and Applications (ICMLA).*

14. **Ahatsham Hayat**, Bilal Khan, Mohammad Hasan, *Enhancing Transfer Learning for Early Forecasting of Academic Performance by Contextualizing Language Models*, *The 19th NAACL Workshop on Innovative Use of NLP for Building Educational Applications (BEA).*

15. **Ahatsham Hayat**, Sharif Wayne Akil, Helen Martinez, Bilal Khan, Mohammad Rashedul Hasan, *Enhancing Zero-Shot Learning of Large Language Models for Early Forecasting of STEM Performance*, *The 2024 ASEE Annual Conference.*

16. **Ahatsham Hayat**, F. Morgado-Dias, T. Choudhury, T. P. Singh, K. Kotecha, *FruitVision: A deep learning based automatic fruit grading system*, *Open Agriculture*, vol. 9, no. 1, 2024, pp. 20220276.

17. **Ahatsham Hayat**, P. Baglat, F. Mendonça, S. S. Mostafa, F. Morgado-Dias, *Machine learning system for commercial banana harvesting*, *Engineering Research Express* 6, 035202.

2023

18. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *The Power of Personalization and Contextualization: Early Student Performance Forecasting with Language Models*, *The 2023 NeurIPS Workshop on Generative AI for Education (GAIED).*

19. H. K. Sharma, H. F. Mahdi, T. Choudhury, **Ahatsham Hayat**, *Automated System Configuration Using DevOps Approach*, *Proceedings of Fourth International Conference on Computer and Communication Technologies*, Lecture Notes in Networks and Systems, vol 606, Springer, Singapore.

20. P. Baglat, **Ahatsham Hayat**, F. Mendonca, A. Gupta, S. S. Mostafa, F. Morgado-Dias, *Non-Destructive Banana Ripeness Detection Using Shallow and Deep Learning: A Systematic Review*, *Sensors*, 2023; 23(2):738.

21. I. V. Manoj, S. Y. Narendranath, P. M. Mashinini, H. Soni, S. Rab, S. Ahmad, **Ahatsham Hayat**, et al., *Artificial neural network-based prediction assessment of wire electric discharge machining parameters for smart manufacturing*, *Paladyn, Journal of Behavioral Robotics*, Vol. 14 (Issue 1), pp. 20220118.

22. **Ahatsham Hayat**, *Breast Cancer Detection System from Thermal Images using SWIN Transformer*, *Journal Press India*, June 15, 2023.

23. V. Shahare, N. Arora, **A. Hayat**, N. Mouje, *Machine Learning Algorithms for Big Data Analytics*, *Demystifying Big Data Analytics for Industries and Smart Societies*. 2023.

24. **A. Hayat**, V. Shahare, A. K. Sharma, N. Arora, *Introduction to Industry 4.0*, *Blockchain and Its Applications in Industry 4.0*. 2023.

25. **A. Hayat**, P. Baglat, F. Mendonça, S. S. Mostafa, F. Morgado-Dias, *Novel Comparative Study for the Detection of COVID-19 Using CT Scan and Chest X-Ray Images*, *International Journal of Environmental Research and Public Health* 20 (2), 1268. 2023.

2022

26. **Ahatsham Hayat**, F. Morgado-Dias, *Deep Learning-Based Automatic Safety Helmet Detection System for Construction Safety*, *Applied Sciences* 12, no. 16: 8268.

27. **Ahatsham Hayat**, F. Morgado-Dias, B. P. Bhuyan, R. Tomar, *Human Activity Recognition for Elderly People Using Machine and Deep Learning Approaches*, *Information* 13, no. 6: 275.

28. N. Silva, G. Machado, I. Viveiros, H. Garcęs, F. Silva, R. Sousa, R. Gouveia, **A. Hayat**, et al., *Agrotech–Sensing Use Cases*, 2022.

2020–2018

29. **Ahatsham Hayat**, S. Jain, V. Shahare, N. Arora, S. Rab, *Real Time Human Activity Recognition Using Smart Phone*, *Advanced Science, Engineering and Medicine*, Volume 12, Number 9, September 2020, pp. 1200–1203(4), American Scientific Publishers.

30. K. Singh, K. Kaushik, **A. Hayat**, V. Shahare, *Role and Impact of Wearables in IoTHealthcare*, *Proceedings of the Third International Conference on Computational Intelligence*. 2020.

31. V. Shahare, N. Arora, **A. Hayat**, N. Mouje, *Smart Home Automation Tool for Energy Conservation*, *International Journal of Recent Technology and Engineering* 8, 5312–5315. 2019.

32. **Ahatsham Hayat**, A. Singh, V. Shahare, N. Arora, *An Efficient System for Early Diagnosis of Breast Cancer using Support Vector Machine*, *International Journal of Engineering and Advanced Technology*. 2019.

33. **Ahatsham Hayat**, N. Arora, K. P. Singh, *An Approach towards Real Time Smart Vehicular System Using Internet of Things*, *International Journal of Research in Engineering, IT and Social Sciences*. 2018.

34. N. Arora, **A. Hayat**, K. Singh, *Achieving Security in Medical Scanned Images Using Extended Image Steganography*, 2018.

Professional Service

- **Organizer**, Madeira International Workshop in Machine Learning, Funchal, Portugal (July 18–22, 2022)
- **Reviewer**, American Society for Engineering Education (ASEE) (2024–2026)
- **Reviewer**, Frontiers in Education Conference (2026)
- **Reviewer**, Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)
- **Reviewer**, Paladyn: Journal of Behavioral Robotics
- **Reviewer**, Signal, Image, and Video Processing (Springer)
- **Reviewer**, International Conference on Advances in Computing, Communication, and Engineering

Honors & Awards

- Chancellor's Graduate Research Travel Award, University of Nebraska–Lincoln (2026)
- Dean's Fellowship, University of Nebraska–Lincoln (2025)
- Milton E. Mohr Fellowship (2024)
- David Swanson Memorial Travel Award (2023)
- Graduate Research Assistantship (2023)
- MHRD Fellowship (2014–2016)

Technical Skills

Programming: Python, PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn
Deep Learning: LLM fine-tuning (QLoRA, LoRA, PEFT), Transformers, CNNs, RNNs, Time-Series Models
Tools: Linux, Git, Docker, Slurm HPC, W&B, Optuna, LaTeX
Domains: NLP, Computer Vision, Behavioral Modeling, Multimodal Learning